



University of Colombo, Sri Lanka

University of Colombo School of Computing

Bachelor of Science in Computer Science

First Year Examination — Semester I - UCSC AY21 [held in Sept./ Oct. 2024]

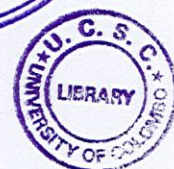
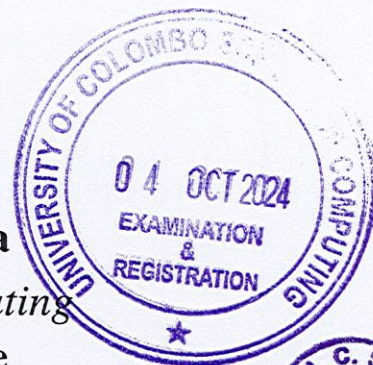
ENH1301 — Application Laboratory

Two (2) Hours

Answer All Questions

Number of Pages = 8

Number of Questions = 4



236

To be completed by the candidate

Index Number

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Important Instructions to candidates

- The students are required to answer all questions within two hours.
- The medium of instructions and questions is English.
- You shall write your answers in **English**.
- This paper has **4** questions on **8** pages.
- Answer **all 4** questions.
- **Write your answers on and only on the space provided** on this question paper.
- Write your index number on each and every page of the question paper.
- Do not tear off any part of this answer book. Under no circumstances may this book (or any part of this book), used or unused, be removed from the Examination Hall by a candidate.
- Questions appear on both sides of the paper. If a page is not printed, please inform the supervisor immediately.
- Any electronic device capable of storing and retrieving text, including electronic dictionaries and mobile phones, are **not allowed**.
- Calculators are **not allowed**.

To be completed by the examiners

1	
2	
3	
4	
Total	

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1. (a). Consider the following directory hierarchy and sequence of commands executed in a Ubuntu GNU/Linux terminal.

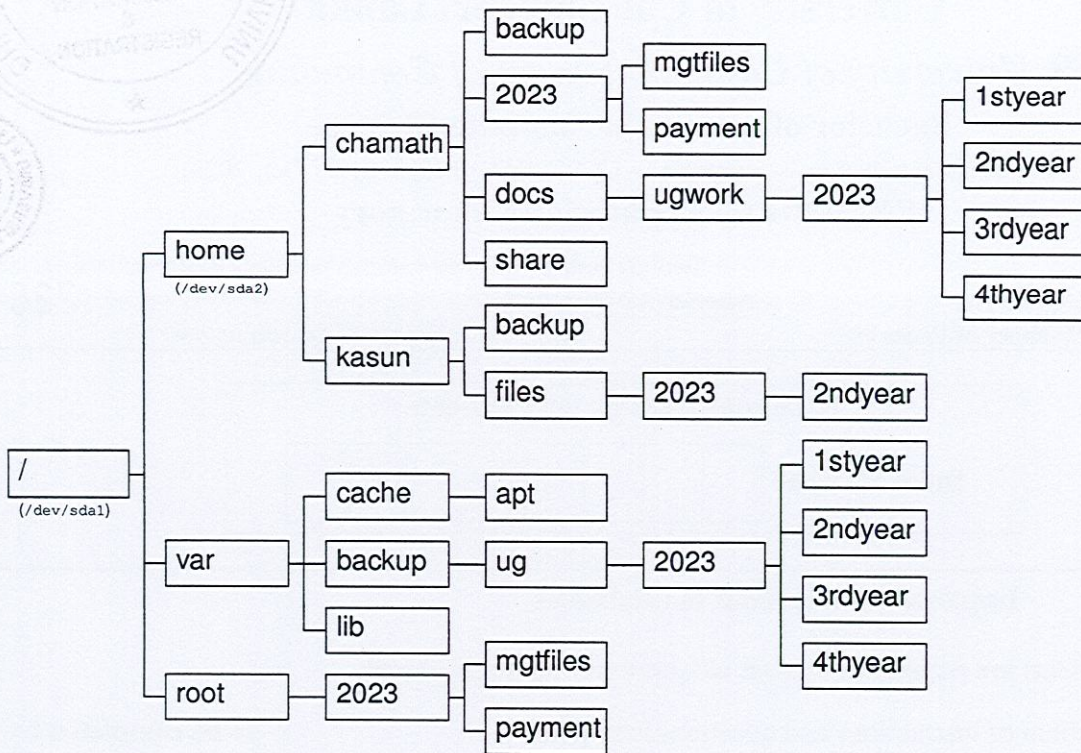


Figure 1: Extract of an Ubuntu GNU/Linux directory structure

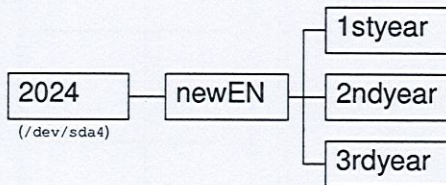


Figure 2: Extract of an Ubuntu GNU/Linux directory structure of /dev/sda4 partition

Assume you logged in as chamath and executed following commands.

```

$ cd docs/ugwork/2023
$ pushd /var/backup/ug/2023
$ echo $?
$ sudo cd /root/2023
$ ls -a ../
$ popd
$ cd ..
$ sudo mount /dev/sda4 /home/chamath/docs/ugwork/2023
$ _____

```

①
②
③
④
⑤
⑥
⑦
⑧
⑨

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10

(11)

(12)

13

14

Use the Figure 1 and sequence of commands given to answer the following sub questions.

[2 marks]

[3 marks]

[2 marks]

[2 marks]

[4 marks]

[2 marks]

[2 marks]

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viii. The command (10) should have been used before (9). Explain why?

[4 marks]

ix. Write the command (12) to create a symbolic link from /var/backup/ug/2023 to a new directory named backup in current directory?

[2 marks]

x. What is the output of the command (14)?

[2 marks]

2. (a). Given below is a US QWERTY keyboard and answer the sub questions using the knowledge of touch typing.

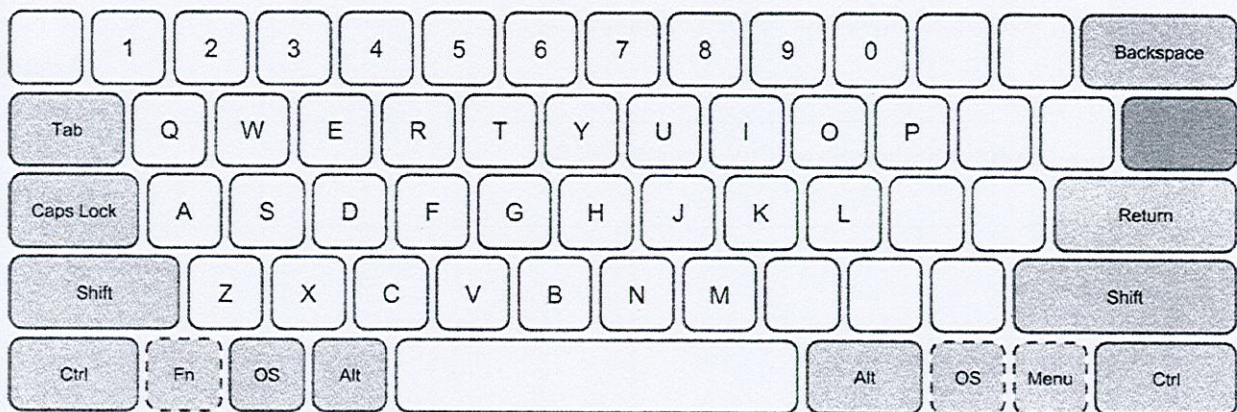


Figure 3: QWERTY Keyboard (credit to Wikipedia, author: Brilliantwiki2)

i. Write letters of **four (04)** keys typed by using **index finger** of your **left hand**.

[4 marks]

ii. Write letters of **three (03)** keys typed by using **ring finger** of your **left hand**.

[3 marks]

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- (b). Assume a GNU/Linux user needs to create an alias `lsl` for the command `ls -l` and use it whenever he/she logs in. What is the file the user should use to create the alias?

[2 marks]

- (c). What is the command a GNU/Linux user should use to get the size of the present working directory?

[3 marks]

- (d). Everything in Linux is a file. The result of `ls -l /home/ruwan/documents/` is given below.

```
-rw-r--r-- 1 ruwan staff 97519 Jun 28 2022 confidential.txt
drwxr-x--- 1 ruwan staff 28148 Sep 30 2022 kamal
-rwxr-x--- 1 ruwan staff 67235 May 31 2022 nimal
lrwxr-xr-x 1 ruwan staff 4096 Sep 30 2022 sudath -> kamal
-r--r--r-- 1 ruwan staff 4254912 Aug 19 2022 sign.sh
```

- i. ruwan executed `chmod 441 kamal`. What are the new permissions of the file `kamal`?

[2 marks]

- ii. After execution of `chmod 441 kamal` what are the new permissions of the file `sudath`?

[2 marks]

- iii. How ruwan can get the same result as after executing `chmod 441 nimal` by using different options instead of `441`?

[3 marks]

- (e). What happens if you execute `!! :p` in a bash shell?

[2 marks]

- (f). `top - 23:03:09 up 4 min, 1 user, load average: 0.75, 0.59, 0.25`
Describe the load average values of the first line of the `top` command output.

[4 marks]

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3. (a). $\text{\LaTeX}$ is a powerful tool for technical writers. Write the output of $\text{\LaTeX}$ script combine with other files given.

<pre> \documentclass[a4paper]{article} \usepackage{listings} \usepackage{natbib} \begin{document} The text \$10_{\log 2}\$ found in the book \citep{kostova2020privacy} \begin{table}[h!] \begin{tabular}{ l c r } \hline {\bf col1} & col2 & col3 \\ \hline col1 & col2 & col3 \\ \hline \end{tabular} \caption{First table} \label{tab:one} \end{table} \lstinputlisting[language=bash]{ sample.sh} \bibliography{ex} \bibliographystyle{apalike} \end{document} </pre>	<pre> #!/bin/bash count=1 while [\$count -le 5] do echo "Count: \$count" ((count++)) done </pre>	<pre> @article{ kostova2020privacy, author = {Li, Tianshi and Agarwal, Yuvraj .}, title={Privacy Engineering}, year={2020}, publisher = {Computing Machinery}, volume = {2}, number = {4}, } </pre>
ex.tex	sample.sh	ex.bib

[20 marks]

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- (b). Write the commands with parameters passed in correct sequence to successfully compile the above $\text{\LaTeX}$ document to get the ex.pdf file.

[5 marks]

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4. (a). Write the key or key combinations or commands to achieve the given tasks in a vim editor command mode.

- i. Move the cursor to the beginning of a line.

[2 marks]

- ii. Copy the current line.

[2 marks]

- iii. Move the cursor 120 lines down.

[2 marks]

- iv. Divide the vim editor view vertically and open a new file named 'Banda'.

[2 marks]

- v. Get the output of the `ls -l` bash shell command into the text file you are working.

[4 marks]

- vi. Navigate forward to the next spelling mistake in vim spellchecker.

[2 marks]

- vii. Navigate to the top of the document.

[2 marks]

- viii. Undo the last modification to the document.

[2 marks]

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(b). Write the **three (03)** areas defined in **Git**.

[3 marks]

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(c). What is the use of `git diff` command?

[2 marks]

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(d). What is the use of `git log` command?

[2 marks]

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